

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design and development of **cross-platform system integrating** wearable data for 100+ users, enabling scalable **data ingestion** and analytics

• Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%

• Built production-ready Python pipeline for automated report generation and scheduled distribution, improving data processing accuracy by 43% and reducing manual reporting overhead

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by **41%**

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings

• Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval

• Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements

• Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure, trends, and performance issues

• Wrote and troubleshoot SQL queries to map operational dependencies, prep data for reporting, and detect quality issues, translating findings clearly for business stakeholders and accelerating root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained 20+ Power BI and Tableau dashboards across Sales, Finance, and Operations using SQL and DAX, supporting KPI reporting, data warehouse queries, and stakeholder decision-making

• Automated recurring report workflows and distributions using VBA and Python, implementing validation and scheduling logic, reducing report preparation time by 42% and improving data consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of **6** to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions

• Tracked sprint delivery and QA issues through JIRA workflows, peer reviewing team work and supporting report promotion from development through testing to production

PROJECTS

FMCG Sales Demand Forecasting

March 2025

View Project

• Analyzed FMCG **sales demand data** across categories and locations to identify trends, seasonal patterns, and key drivers impacting inventory

• Built and compared forecasting models in R including ARIMA, SARIMA, and regression, using ggplot2 for visualization and evaluating demand patterns with RMSE and MAPE metrics

• Derived actionable insights on promotions, pricing, and stock behavior, enabling better inventory planning and reducing stockouts and overstocking risks